

Project Title
Unicode Pattern Search Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Unicode Pattern Search Engine is a text retrieval system designed to search and identify patterns across multilingual text using the Unicode standard. Unlike traditional search engines that primarily focus on ASCII or language-specific character sets, this system supports a wide range of global languages and scripts, enabling accurate and efficient text processing. The search engine utilizes Unicode encoding to represent characters consistently, ensuring compatibility with diverse languages such as English, Telugu, Hindi, Japanese, Arabic, and many others. Advanced pattern-matching techniques are employed to improve search accuracy, reduce ambiguity, and deliver relevant results even when handling complex character combinations. The proposed system is suitable for applications such as digital libraries, multilingual databases, search portals, document management systems, and language processing tools. By providing fast, reliable, and language-independent search capabilities, the Unicode Pattern Search Engine enhances information retrieval and accessibility for users worldwide. This project demonstrates the importance of Unicode-based text processing in building modern search systems that effectively support global communication and multilingual computing environments.

Signature of the Guide